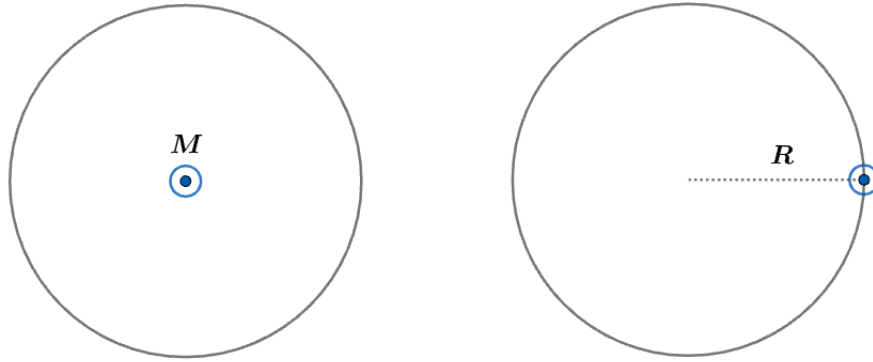


2008 F=ma Exam: Problem 12

Kevin S. Huang



For a disk rotating around its center, we have

$$E = \frac{1}{2} I_{\text{disk}} \omega^2$$

For a disk rotating around its edge, we have

$$E' = \frac{1}{2} I_{\text{edge}} \omega^2$$

Recall the moment of inertia of a disk is given by

$$I_{\text{disk}} = \frac{1}{2} M R^2$$

By the parallel axis theorem, a disk rotating around its edge has moment of inertia

$$I_{\text{edge}} = I_{\text{disk}} + M R^2 = \frac{3}{2} M R^2$$

Thus,

$$\frac{E'}{E} = \frac{I_{\text{edge}}}{I_{\text{disk}}} = 3$$

so the answer is D.